

CO4 ASSESSMENT 3

Q1. Supermarket Purchases – Customer Segmentation

The dataset contains five customers based on **Customer Visits/Month, Average Spend, and Items Bought**. The question asks us to propose a clustering approach for customer segmentation and explain how it supports targeted marketing.

Answer

Clustering technique: K-Means Clustering

K-Means is suitable because it groups customers with similar purchasing behavior into different clusters.

Step 1: Select the features

Use these three attributes:

- Customer Visits/Month
- Average Spend (₹)
- Items Bought

Step 2: Preprocess the data

The values have different ranges, so **feature scaling/standardization** should be performed before applying K-Means.

Step 3: Apply K-Means

Choose **K = 3** to create three customer segments.

Based on the given data, the customers can be interpreted as:

- **Cluster 1 – Frequent Customers:** C1, C3, C5
They visit the supermarket frequently, spend around ₹1,800–₹2,100, and buy many items.
- **Cluster 2 – High-Spending Customers:** C2
This customer visits less frequently but has a high average spending of ₹5,000.
- **Cluster 3 – Occasional High-Spending Customers:** C4

This customer visits very few times but has the highest average spending of ₹6,000.

Step 4: Targeted Marketing

The supermarket can use these clusters to provide personalized offers:

Customer Segment	Marketing Strategy
Frequent Customers	Loyalty points, membership rewards and regular discounts
High-Spending Customers	Premium products, exclusive offers and personalized recommendations
Occasional Customers	Coupons, special discounts and offers to encourage frequent visits

Advantages

- Identifies customers with similar purchasing patterns.
- Helps the supermarket understand customer behavior.
- Enables personalized marketing campaigns.
- Improves customer satisfaction and retention.
- Can increase sales by targeting the right customers.

Conclusion

K-Means clustering can effectively segment supermarket customers according to their visits, spending, and number of items purchased. These segments allow the supermarket to provide **targeted and personalized marketing offers**, resulting in better customer engagement and improved sales.

Q2. Credit Card Transactions – Fraud Detection

The dataset contains **Transaction Amount, Location, and Time** for five transactions. The task is to design a clustering-based solution to identify unusual or fraudulent transactions.

Answer

Clustering technique: K-Means Clustering

K-Means can group transactions having similar characteristics. Transactions that are very different from the normal groups can be treated as **potentially suspicious transactions**.

Step 1: Select the features

Use:

- Transaction Amount (₹)
- Location
- Time (hrs)

Since **Location is categorical**, it should first be converted into numerical form, such as **one-hot encoding**.

Step 2: Preprocess the data

The transaction amount and time have different scales. Therefore, numerical features should be **standardized** before clustering.

Step 3: Apply K-Means Clustering

Choose a suitable value of **K**, for example **K = 2**:

- **Normal transaction cluster**
- **Unusual transaction cluster**

From the given data:

Transaction Amount (₹) Location Time Observation

T1	200	Chennai	10	Normal pattern
T2	5000	Delhi	2	Unusual
T3	150	Chennai	11	Normal pattern

T4	7000	Mumbai	1	Unusual
T5	180	Chennai	12	Normal pattern

The transactions **T2 and T4** have much higher amounts and occur at unusual times compared with T1, T3 and T5. Therefore, they can be considered **potentially suspicious**.

Step 4: Fraud Detection

After clustering:

- Transactions close to the **normal cluster** → likely normal.
- Transactions far from the normal cluster → **potentially fraudulent/unusual**.
- A distance threshold can be used to flag suspicious transactions for further investigation.

Step 5: Practical Use

A bank can use this system to:

- Detect unusual spending patterns.
- Flag suspicious transactions.
- Send alerts for further verification.
- Reduce financial losses due to fraudulent activity.
- Improve transaction monitoring.

Conclusion

K-Means clustering can group credit card transactions according to their amount, location, and time. Transactions that significantly differ from the normal cluster can be flagged as **potentially fraudulent** and investigated further. This provides an effective clustering-based approach for credit card fraud detection.

Q3. Telecom Usage – Identifying Churn-Prone Users

The dataset contains **Call Duration, Data Usage, and Recharge** information for five telecom users. The task is to explain how clustering can identify churn-prone users and improve customer retention.

Answer

Clustering technique: K-Means Clustering

K-Means can group telecom users according to their usage and recharge behavior. Users with similar behavior will belong to the same cluster.

Step 1: Select the features

Use:

- Call Duration (minutes)
- Data Usage (GB)
- Recharge (₹)

Step 2: Preprocess the data

Since the three features have different ranges, **standardization** should be performed before applying K-Means.

Step 3: Apply K-Means

Choose **K = 3** to create three user groups:

- **High-usage users**
- **Medium-usage users**
- **Low-usage users**

From the given dataset:

User Call Duration Data Usage Recharge User Type

U1	300 min	5 GB	₹200	High usage
U2	100 min	2 GB	₹100	Low usage
U3	350 min	6 GB	₹220	High usage

U4	80 min	1 GB	₹90	Low usage
U5	320 min	5.5 GB	₹210	High usage

The **low-usage users (U2 and U4)** can be considered more likely to reduce their usage or leave the service, so they can be treated as **potential churn-prone users**. This is an interpretation based on the supplied dataset; the dataset alone does not prove that they will actually churn.

Step 4: Retention Strategies

The telecom company can:

- Give special recharge offers to low-usage users.
- Provide additional data or call benefits.
- Send personalized discounts.
- Encourage users to choose suitable plans.
- Monitor changes in usage over time.

Step 5: Benefits

- Identifies different types of telecom users.
- Helps detect users with low engagement.
- Enables personalized offers.
- Improves customer satisfaction.
- Helps reduce customer churn.

Conclusion

K-Means clustering can group telecom users based on their call duration, data usage, and recharge amount. Low-usage groups can be monitored and provided with **personalized offers and suitable plans**, helping the telecom company improve customer retention.

Q4. E-commerce Behavior – User Grouping

The dataset contains **Product Views, Clicks, and Purchases** for five users. The task is to suggest a clustering technique and explain how it can improve recommendation systems.

Answer

Clustering technique: K-Means Clustering

K-Means can group users who have similar browsing, clicking, and purchasing behavior. These groups can then be used to provide personalized product recommendations.

Step 1: Select the features

Use:

- Product Views
- Clicks
- Purchases

Step 2: Preprocess the data

The features have different numerical ranges, so **standardization** should be performed before applying K-Means.

Step 3: Apply K-Means

Choose **K = 3** to create three user groups:

- **Highly active users**
- **Moderately active users**
- **Less active users**

Based on the given dataset:

	Product Views	Clicks	Purchases	User Type
User				
U1	50	20	5	Highly active
U2	30	10	2	Moderately active
U3	60	25	6	Highly active
U4	20	8	1	Less active

U5	55	22	5	Highly active
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Users **U1, U3, and U5** show high engagement, while **U4** shows relatively low engagement.

Step 4: Improve Recommendation Systems

The e-commerce platform can use the clusters to:

- Recommend products based on the behavior of similar users.
- Show frequently viewed products to highly active users.
- Recommend related products to users with purchasing activity.
- Provide personalized offers to less-active users.
- Improve the relevance of product recommendations.

Benefits

- Identifies groups of users with similar behavior.
- Provides personalized recommendations.
- Improves customer engagement.
- Can increase product purchases.
- Helps understand customer preferences.

Conclusion

K-Means clustering can group e-commerce users based on their product views, clicks, and purchases. The identified user groups can be used to provide **personalized product recommendations**, improving user engagement and the effectiveness of the recommendation system.

Q5. Traffic Data – Identifying Congestion Zones

The dataset contains **Location, Vehicle Count, Average Speed, and Delay** for five locations. The task is to design a clustering-based solution to identify traffic congestion zones and explain its use in traffic management.

Answer

Clustering technique: K-Means Clustering

K-Means can group traffic locations with similar congestion characteristics based on vehicle count, average speed, and delay.

Step 1: Select the features

Use:

- Vehicle Count
- Average Speed (km/h)
- Delay (minutes)

Step 2: Preprocess the data

Since the features have different ranges, **standardization** should be performed before applying K-Means.

Step 3: Apply K-Means

Choose **K = 3** to create three groups:

- **Low congestion**
- **Medium congestion**
- **High congestion**

Based on the given dataset:

Location	Vehicle Count	Avg Speed	Delay	Congestion Level
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L1	200	30 km/h	10 min	Low
L2	500	15 km/h	25 min	High
L3	220	28 km/h	12 min	Low
L4	600	10 km/h	30 min	High
L5	210	32 km/h	9 min	Low

Locations **L2 and L4** have high vehicle counts, low average speeds, and high delays, indicating **high congestion**. Locations **L1, L3, and L5** show comparatively lower congestion.

Step 4: Traffic Management

The traffic department can use the clusters to:

- Identify highly congested zones.
- Adjust traffic signal timings.
- Divert vehicles to less congested routes.
- Deploy traffic police or emergency services.
- Improve road infrastructure in heavily congested areas.
- Provide real-time traffic information to drivers.

Benefits

- Identifies congestion-prone locations.
- Helps in better traffic planning.
- Reduces travel delays.
- Supports efficient road management.
- Helps authorities make data-driven decisions.

Conclusion

K-Means clustering can group traffic locations based on vehicle count, average speed, and delay. The identified **high-congestion zones** can then be targeted with appropriate traffic-control measures, helping to reduce congestion and improve overall traffic management.